

Homework — 1.3.3 Networks

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.3.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Complete the table by naming the **TCP/IP stack layer** described in each row. [2] (AO1)

Description	Layer
Interacts with the user's software, e.g. a browser using HTTP to request a web page	
Establishes the end-to-end connection and splits the data into packets, adding sequence numbers	
Adds the source and destination IP addresses so routers can route each packet	
Moves the data over the physical network, addressing the actual hardware using MAC addresses	

2. Write **True** or **False** for each statement. If a statement is false, write a correction in the final column. [4] (AO1)

Statement	True / False	Correction if false
A LAN covers a small geographical area, such as a single site.		
In packet switching, every packet must follow the same route to the destination.		
Standard protocols allow devices made by different manufacturers to communicate successfully.		
The internet is an example of a LAN.		

3. A user types a URL into their browser and presses Enter. Number the steps from 1 to 6 in the order they happen. [3] (AO2)

Step	Order (1–6)
The DNS server resolves the domain name to an IP address and returns it.	
The browser sends a request to the web server at that IP address.	
The user types the URL into the browser and presses Enter.	
The packets are reassembled in order using their sequence numbers and the page is displayed.	
The browser sends the domain name to a DNS server.	
The server's response is split into packets, which may be routed independently across the internet.	

4. A company sets up a **client-server** network instead of a peer-to-peer network. Give **two** advantages and **one** disadvantage of this choice. [3] (AO2)

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5. The company also installs a **firewall**. Describe what the firewall does to help protect the network. [2] (AO1)

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Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.2) State what is meant by **referential integrity** in a relational database. [2] (AO1)

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7. (1.3.1) State **two** properties that a good hash function should have. [2] (AO1)

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8. (1.2.2) Explain **one** task carried out by a **linker** when an application is built. [2] (AO1)

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